

Q1. Given below are two statements:

NEET - 2024

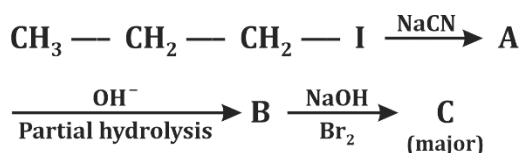
Statement I : Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II : Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the *correct answer* from the option given below :

- Both Statement I and Statement II are false.
- Statement I is correct but Statement II is false.
- Statement I is incorrect but Statement II is true.
- Both Statement I and Statement II are true.

Q2. Identify the major product C formed in the following reaction sequence: NEET - 2024



- butylamine
- butanamide
- α - α - bromobutanoic acid
- propylamine

Q3 Match List-I with List-II

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	List-I (Amine)		List-II (pK_β value)(pK_β value)
A.	Benzenamine	I.	3.38
B.	Phenylmethanamine	II.	9.38
C.	Methenamine	III.	4.70
D.	N-Ethylethanamine	IV.	3.0

Choose the correct answer from the options given below:

- A-II, B-III, C-I, D-IV

2. A-III, B-II, C-I, D-IV
3. A-II, B-I, C-III, D-IV
4. A-I, B-III, C-II, D-IV

Q4. Match List-I with List-II

NEET - 2024

	List-I (Reagent)		List-II (name of the reaction)
A.	Carbylamine test	I.	phenol
B.	Bayer's test	II.	acetone
C.	Iodoform test	III.	ethylene
D.	Phthalein dye test	IV.	aniline

Choose the correct answer from the options given below:

1. A-IV, B-III, C-II, D-I
2. A-IV, B-II, C-III, D-I
3. A-IV, B-II, C-I, D-III
4. A-II, B-III, C-I, D-IV

Q5. Given below are two statements:

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Statement I:	Benzendiazonium chloride is a colourless crystalline solid. It is insoluble in water but reacts with water when warmed.
Statement II:	Benzendiazonium chloride on reacting with HCl in presence of copper powder gives chlorobenzene as the product. This is an example of Gatterman reaction.

In the light of the above statements, choose the correct answer from the option given below:

1. Both **Statement I** and **Statement II** is true.
2. Both **Statement I** and **Statement II** is false.
3. **Statement I** is true but **Statement II** is false.
4. **Statement I** is false but **Statement II** is true.

Q6. The compound that does not undergo Friedel-Crafts alkylation reaction but gives a positive carbylamine test is:

NEET - 2024

1. aniline
2. pyridine



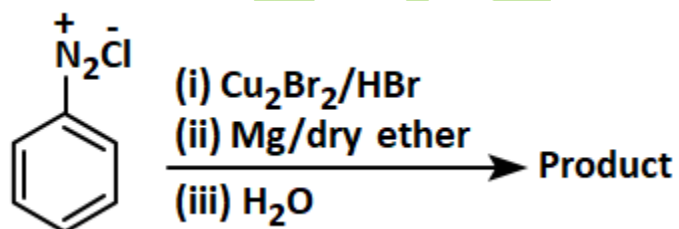
3. N-methylaniline

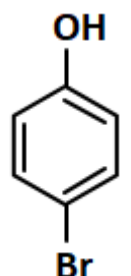
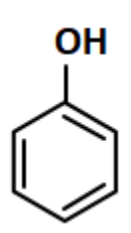
4. triethylamine

Q7. Among the given reactions, which one does not result in the formation of a primary amine as the product? **NEET - 2023**

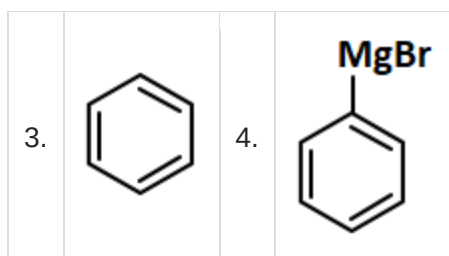
1.	$\text{CH}_3\text{CONH}_2 \xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) LiAlH}_4} \text{Product}$
2.	$\text{CH}_3\text{CONH}_2 \xrightarrow{\text{Br}_2/\text{KOH}} \text{Product}$
3.	$\text{CH}_3\text{CN} \xrightarrow[\text{(ii). H}_3\text{O}^+]{\text{(i). LiAlH}_4} \text{Product}$
4.	$\text{CH}_3\text{NC} \xrightarrow[\text{(ii). H}_3\text{O}^+]{\text{(i). LiAlH}_4} \text{Product}$

Q8. Identify the product in the following reaction: **NEET - 2023**



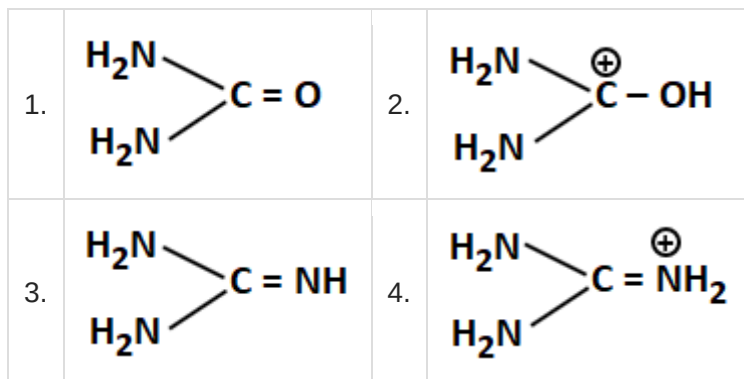
1.		2.	
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Q9. The least basic compounds/species among the following is:

NEET – 2023



Q10. Choose the correct sequence of reagents in the conversion of 4-nitrotoluene to 2-bromotoluene.

NEET – 2023

1. NaNO_2/HCl ; Sn/HCl ; Br_2 ; $\text{H}_2\text{O}/\text{H}_3\text{PO}_2$
2. Sn/HCl ; NaNO_2/HCl ; Br_2 ; $\text{H}_2\text{O}/\text{H}_3\text{PO}_2$
3. Br_2 ; Sn/HCl ; NaNO_2/HCl ; $\text{H}_2\text{O}/\text{H}_3\text{PO}_2$
4. Sn/HCl ; Br_2 ; NaNO_2/HCl ; $\text{H}_2\text{O}/\text{H}_3\text{PO}_2$

Q11. Given below are two statements:

NEET – 2022

I	Primary aliphatic amines react with HNO_2 to give unstable diazonium salts.
II	Primary aromatic amines react with HNO_2 to form diazonium salts which are stable even above 300 K.

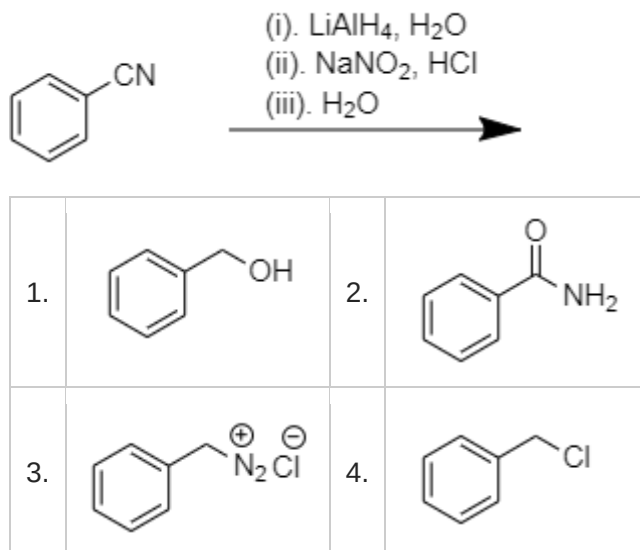
In the light of the above statements, choose the most appropriate answer from the options given below:

- | | |
|----|----------------------------------|
| 1. | I is incorrect but II is correct |
| 2. | Both I and II are correct |
| 3. | Both I and II are incorrect |



4. I is correct but II is incorrect

Q12. The product formed from the following reaction sequence is **NEET - 2022**



Q13 Match list-I with list-II:

NEET - 2022

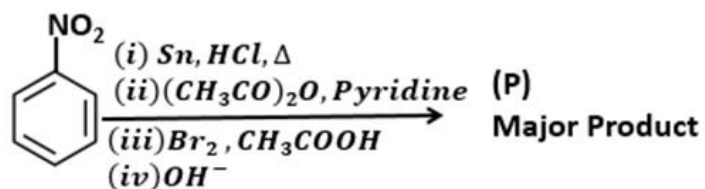
	List-I (Amines)		List-II (pK_b values)
(a)	N-methylmethanamine	(i)	9.30
(b)	Ammonia	(ii)	9.38
(c)	N-methylaniline	(iii)	4.75
(d)	Benzenamine	(iv)	3.27

Choose the correct answer from the options given below:

	(a)	(b)	(c)	(d)
1.	(iv)	(ii)	(i)	(iii)
2.	(iv)	(iii)	(i)	(ii)
3.	(iii)	(iv)	(i)	(ii)
4.	(i)	(iv)	(iii)	(ii)



Q14 The major product (P) formed in the following reaction sequence is : **NEET - 2022**



1.		2.	
3.		4.	

Q15. Match column I (Reaction) with column II (Product formed) and mark the appropriate choice: **NEET - 2022**

	Column - I (Reaction)		Column - II (Product formed)
(a)	Gabriel synthesis	(i)	Benzaldehyde
(b)	Kolbe synthesis	(ii)	Ethers
(c)	Williamson synthesis	(iii)	Primary amines
(d)	Etard reaction	(iv)	Salicylic acid

	(a)	(b)	(c)	(d)
1.	(iii)	(i)	(ii)	(iv)

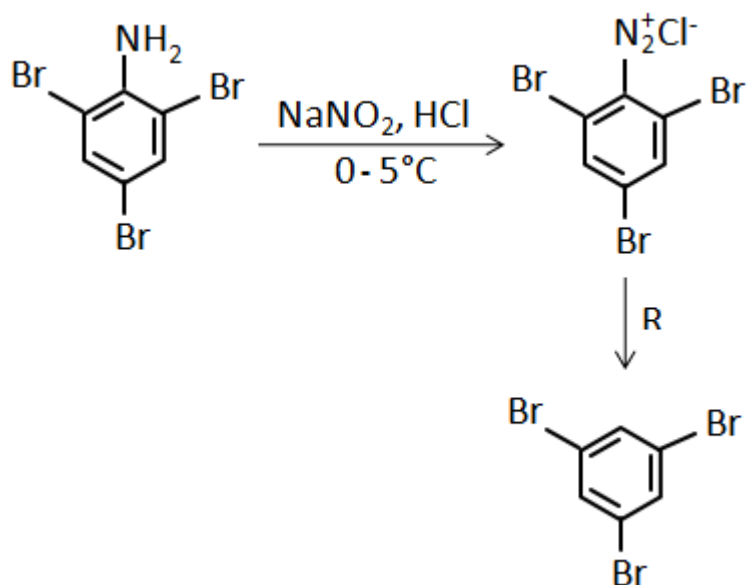


2.	(ii)	(iii)	(i)	(iv)
3.	(iv)	(iii)	(i)	(ii)
4.	(iii)	(iv)	(ii)	(i)

Q16. Identify the compound that will react with Hinsberg's reagent to give a solid which dissolves in alkali. **NEET - 2022**

1.	$\text{H}_3\text{C}-\text{CH}_2-\text{NH}_2$	2.	$\text{H}_3\text{C}-\text{CH}_2-\text{N}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$
3.	$\text{H}_3\text{C}-\text{CH}_2-\text{NO}_2$	4.	$\text{H}_3\text{C}-\text{CH}_2-\text{NH}-\text{CH}_3$

Q17. The reagent 'R' in the given sequence of a chemical reaction is : **NEET - 2021**

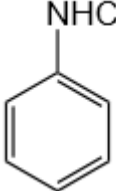
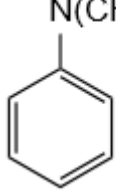
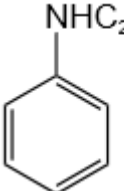
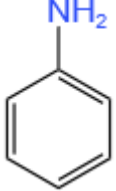


1.	HI	2.	CuCN/KCN
3.	H ₂ O	4.	CH ₃ CH ₂ OH



Q18. Which amine gives the carbylamine test?

NEET – 2020

1.		2.	
3.		4.	

Q19. The reaction of propanamide with ethanolic sodium hydroxide and bromine will give:

NEET – 2020

1.	Ethylamine	2.	Methylamine
3.	Propylamine	4.	Aniline

Q20. What is the correct order of the methyl-substituted amines' basic strength in an aqueous solution?

NEET – 2019

1. $\text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH} > (\text{CH}_3)_3\text{N}$

2. $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_3\text{N}$

3. $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$

4. $(\text{CH}_3)_3\text{N} > \text{CH}_3\text{NH}_2 > (\text{CH}_3)_2\text{NH}$

Q21. Which amine reacts with Hinsberg's reagent to produce an alkali insoluble product?

NEET – 2019



1.	$ \begin{array}{c} \text{H}_3\text{C}-\text{H} \\ \diagup \quad \diagdown \\ \text{C}-\text{NH}-\text{C} \\ \diagdown \quad \diagup \\ \text{H}_3\text{C} \quad \text{H}-\text{CH}_3 \end{array} $	2.	$ \begin{array}{c} \text{H}_3\text{C} \quad \text{CH}_2 \quad \text{H}_2 \\ \diagup \quad \diagdown \quad \diagup \\ \text{H}_3\text{C}-\text{C}-\text{N}-\text{C}-\text{CH}_3 \\ \quad \quad \quad \diagdown \\ \quad \quad \quad \text{H}_2 \end{array} $
3.	$ \begin{array}{c} \text{NH}_2 \quad \text{H}_2 \\ \diagup \quad \diagdown \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \diagdown \quad \diagup \\ \text{H}_3\text{C} \quad \text{H}_2 \end{array} $	4.	$ \begin{array}{c} \text{O} \\ \parallel \\ \text{H}_3\text{C}-\text{C}-\text{C}-\text{CH}_3 \\ \quad \quad \diagup \quad \diagdown \\ \quad \quad \text{H}_2\text{C}-\text{CH}_3 \quad \text{NH}_2 \end{array} $



ANSWER KEY

1. Ans. (4)
2. Ans. (4)
3. Ans. (1)
4. Ans. (1)
5. Ans. (4)
6. Ans. (4)
7. Ans. (4)
8. Ans. (3)
9. Ans. (2)
10. Ans. (1)
11. Ans. (4)
12. Ans. (1)
13. Ans. (3)
14. Ans. (4)
15. Ans. (4)
16. Ans. (1)
17. Ans. (4)
18. Ans. (4)
19. Ans. (1)
20. Ans. (2)
21. Ans. (1)
22. Ans. (4)
23. Ans. (1)
24. Ans. (1)
25. Ans. (3)
26. Ans. (3)
27. Ans. (3)
28. Ans. (1)

